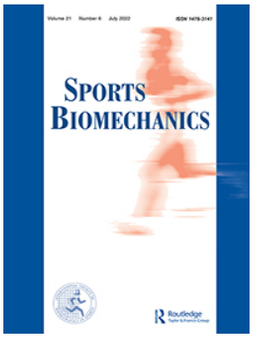

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Do healthy athletes exhibit at-risk biomechanics for anterior cruciate ligament injury during pivoting movements?

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ABSTRACT

A consistent injury mechanism involving multiple joints has been highlighted in athletes experiencing anterior cruciate ligament (ACL) rupture. However, if and to what extent healthy athletes incur an unsafe biomechanical profile during high-dynamics movements is unknown. The present study aimed to investigate the occurrence of the ACL risk profile in a competitive pivoting sports population.Thirty-four athletes (22.8 ± 4.1 y) performed a frontal deceleration and a change of direction at 90°. Full-body kinematics was collected through 15 wearable inertial sensors (Awinda, Xsens). Nine ACL risk factors were defined based on four categories: limited lower limb flexion, valgus collapse, foot rotation and trunk rotation. A movement trial was considered 'at-risk' in the presence of at least 5 simultaneous risk factors. The rate of athletes with at-risk movements was assessed and multivariate regression for associated outcomes was conducted.The overall rate of injury profile occurrence was 9–12%. The injury profile was identified at least in one trial in 24 athletes (71%) and three trials in 5 athletes (15%). Significant associations were found for higher approaching speed (OR = 4.3) and female sex (OR = 4.8). A large occurrence of the typical ACL injury biomechanical profile was noticed. Large screenings are advisable to identify at-risk athletes and promote preventative strategies.

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ACL; wearables; injury mechanism; injury prevention; football

Introduction

In the last decade, the anterior cruciate ligament (ACL) injury rate has doubled in the young competitive population involved in pivoting sports (Weitz et al., 2020). The ACL injury has devastating consequences on young athletes' career: the return to sport at pre-injury level is often compromised and the sport participation significantly drops (Ithurburn et al., 2019), with both short- and long-term decay in physical and psychological fitness (Ithurburn et al., 2019; Rothrauff et al., 2020). Since non-contact and

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indirect-contact ACL injury has been noticed to be the most frequent (Della Villa et al., 2020) and indicative of re-ruptures (Della Villa et al., 2021), the preventative training strategies have been increasingly connected to movement quality enhancement.

Extensive work regarding the identification of biomechanical risk factors for ACL injury has been conducted (King et al., 2021a, 2021b). Knee rotations, trunk position, foot position and limited lower limb flexion have been identified as major risk factors (Bates et al., 2020; Donelon et al., 2020; Paterno et al., 2015). However, the assessment of the single risk factors has been poorly transferred to daily clinical practice (Donnell-Fink et al., 2015) and a more comprehensive approach has been advocated (Hewett & Bates, 2017). Recently, several authors suggested that the ACL rupture is caused by a recurring combination of multiple risk factors (Della Villa et al., 2020; Lloyd, 2021; Lucarno et al., 2021). Lloyd defined such a biomechanical profile as a 'perfect ACL loading storm' (Lloyd, 2021). Three extensive video-analysis studies precisely depicted the entire body pose at the frame of ACL injury in professional male and female football and male rugby players, providing quantitative information (Della Villa et al., 2020, 2021; Lucarno et al., 2021).

The use of quantitative multi-joint cut-offs allows the clinically relevant identification of anomalies during motion. Currently, there is little knowledge on if and how frequent the healthy athletes experience a motion profile close to the ACL injury mechanisms during sport-specific movements. A large occurrence at group or individual level would raise awareness of the need for large screenings and dedicated training actions already in the youth.

Thus, the aim of the present study was to investigate the occurrence of biomechanical risk profile for ACL injury in a competitive healthy population during the execution of high-dynamics movements. It was hypothesised that (I) the ACL injury profile frequently arises in healthy athletes' movements, that (II) female sex and movement speed are associated with the occurrence of the injury profile and that (III) knee joint loading is higher in subjects showing the injury profile.

Materials and methods

Participants

Thirty-four competitive young, healthy athletes (22.8 ± 4.1 years, 18 males and 16 females, Tegner Level 9) were enrolled in the study. All athletes belonged to pivoting field sports (soccer: n = 28, rugby: n = 4, basketball: n = 2) with frequent pivoting movements. Athletes with previous severe musculoskeletal injuries (>28-day recovery time), previous lower limb surgery, BMI >35 and cardiopulmonary or cardiovascular disorders were excluded from the study (Renshaw & Goodwin, 2016; Tourny et al., 2014). All athletes signed informed consent before starting the acquisition protocol. The study received IRB approval by the Institutional Review Board of Area Vasta Emilia Romagna Centro (IRB approval: 555/2018/Sper/IOR of 12/09/2018) and was registered on ClinicalTrials.gov (Identifier: NCT03840551). An a priori power analysis was conducted in G*Power (v3.1, Brunsbüttel, Germany) to determine the adequate sample size (Al-Amri et al., 2018). At least 28 athletes were required to have a power of 0.9 with a minimum effect size of 0.5 and a type I error of 0.05.

Data collection

Data were collected in a specialised laboratory that was equipped with artificial turf at the Green Room of the Isokinetic Medical Center of Bologna (Italy). Every athlete performed two motor tasks: a frontal sprint with deceleration (DEC), and a pre-planned change of direction at 90° (COD) (Figure 1). The DEC task consisted of a frontal sprint followed by a stable stop in a single support, while the COD consisted of a frontal sprint with sidestep cutting at 90°. The present tasks resemble two of the most frequent dynamics in pivoting sports (i.e., pressing and cuts) and are widely recognised as situational patterns for non-contact ACL injury occurrence (Della Villa et al., 2020; Grassi et al., 2017; Lucarno et al., 2021). The athletes were instructed to 'perform the task at their best' (i.e., sprint as fast as they could). Before data collection, a short warm-up and a few unrecorded repetitions were carried out to gain confidence with the environment and the movements. Three valid repetitions per leg of each task were recorded. A sports medicine physician specialised in sports biomechanics (FDV) instructed each athlete and verified the trials' validity.

A set of 15 wearable inertial sensors (MTw Awinda, Xsens Technologies, Enschede, The Netherlands) placed on upper limbs, trunk and lower limbs was used to capture motion during the execution of the tasks (Di Paolo et al., 2021; Poitras et al., 2019). The

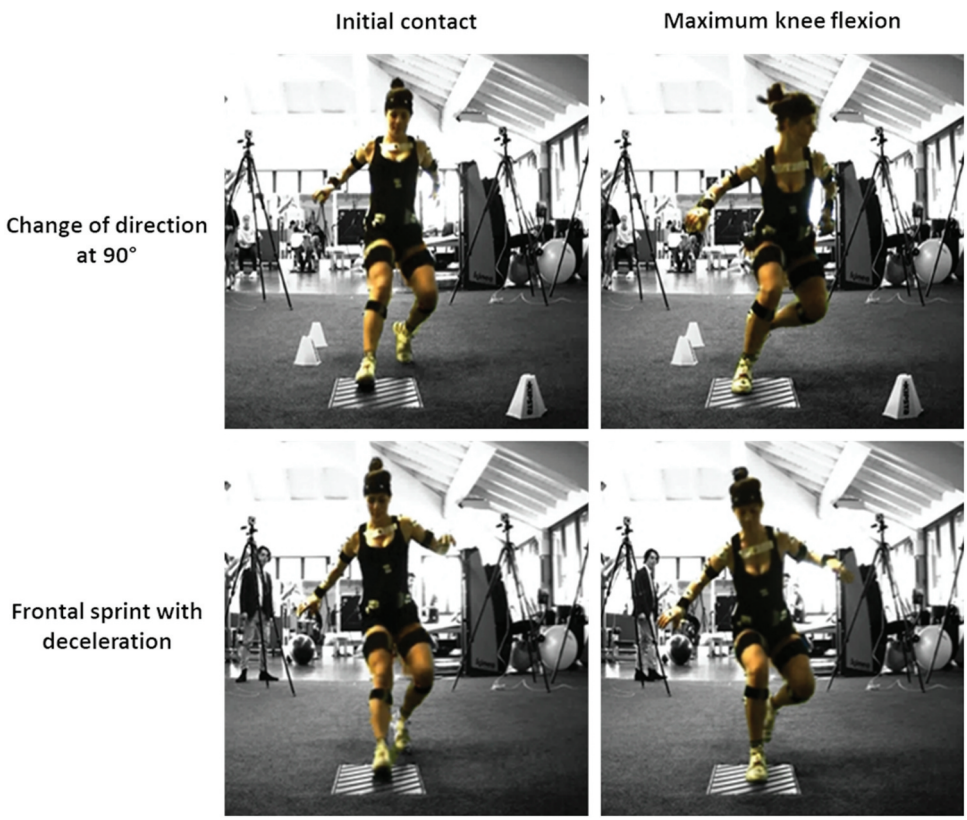

Figure 1. Example of the investigated motor tasks. The athletes' biomechanics during the ultimate foot strike before change of direction was considered in relation to the ACL injury risk profile.

full-body joint kinematics was collected at a sampling frequency of 60 Hz. The sensors placement was performed by a single experienced operator. The system calibration was performed in both static (upright standing) and dynamic (walking) conditions per athlete. A marker-based optoelectronic system (Vicon Vero, Vicon Motion Systems Ltd., Oxford, UK) with 42 retroreflective markers, one force platform and 10 cameras was also used to capture motion. In the present study, the marker-based system was only used to compute joint dynamics. The wearable system and the marker-based system were synchronised for direct time comparison. The reliability of the wearable system in the assessment of high dynamics movements was confirmed in previous studies (Di Paolo et al., 2021; Poitras et al., 2019).

Data processing

The data processing was performed in the dedicated Xsens software and in a custom Matlab script (The MathWorks, Natick, MA, USA). The kinematical data (°) of ankle, knee, hip and trunk joints extracted from the wearable sensors were processed. The ultimate strike before the change of direction was isolated for each task (Dos'Santos et al., 2019; Weir et al., 2019) (Figure 1).

According to the current literature on non-contact ACL injury profile (Bahr, 2005; Della Villa et al., 2020; Donelon et al., 2020; Hewett & Schilaty, 2018; Lucarno et al., 2021), nine biomechanical risk factors were identified across four categories: limited lower limbs flexion (knee, hip, ankle), knee valgus collapse (knee valgus, hip abd/adduction and internal rotation), foot rotation, trunk rotation (ipsilateral bending, contralateral rotation). A detailed description of the biomechanical risk factors defining the injury profile is reported in Table 1. In brief, each risk factor cut-off was chosen on the basis of literature data and, where a specific cut-off was absent, the present cohort's data distribution was used (i.e., first and third quartiles were used). The injury profile was assessed separately per

Table 1. List of the biomechanical risk factors for ACL injury assessed through the wearable sensors with relative thresholds.

Description of the ACL biomechanical risk profile	
Risk factor	Frame (initial contact/maximum knee flexion)
Limited lower limb flexion (°)	
Hip, flexion [a]	<40.3/43.1
Knee, flexion [a]	<20.1/46.2
Ankle, plantarflexion [a]	<−8.5/0.1
Knee valgus collapse	
Knee, valgus[b]	>5.3/5.1
Hip, abd/adduction [c]	<0.4/−2.7; >19.0/19.7
Hip, internal rotation[b]	>15.1/13.0
Foot rotation	
Ankle, internal/external rotation [c]	<−15.4/-16.0; >−0.1/-1.1
Trunk bending	
Trunk, ipsilateral bending [c,d]	>14.5/22.9
Trunk, contralateral rotation [c,d]	>5.8/11.9

a = threshold adapted from literature (Della Villa et al., 2020; Lucarno et al., 2021) with 15% increase; b = referred to the third quartile of the present study data distribution; c = referred to the first and third quartiles of present study data distribution (i.e., values out of the interquartile range); d = with respect to the striking foot.

each single movement trial. A movement trial was considered 'at-risk' in the presence of at least five out of nine simultaneous risk factors. This way, in each at-risk movement trials, unsafe biomechanics was defined as a combination of multiple risk factors. In particular, an at-risk trial included a risk factor in at least two anatomical planes out of three (e.g., one on frontal and one on sagittal plane), two joints out of four (e.g., one on hip and one on knee) and two categories out of four (e.g., one related to the knee valgus collapse and one related to the limited lower limb flexion). Risk factors were assessed at initial foot contact (IC) and at the maximum knee flexion angle (MKF), separately. Both frames have been investigated in biomechanical studies on non-contact ACL injury prevention and rehabilitation (Dos'Santos et al., 2019; Weir et al., 2019).

Furthermore, the knee abduction moment (KAM), knee flexion moment (KFM) and knee rotation moment (KRM) were extracted from the marker-based system and normalised by each athlete's body weight. As highlighted in previous prospective studies (Bates et al., 2019; Hewett et al., 2015; Sigurðsson et al., 2020), the knee joint moments are widely recognised risk factors for non-contact ACL injury. In the present study, the knee joint moment waveforms were used to corroborate the effectiveness of the biomechanical injury profile definition here proposed towards the knee loading and the ACL injury risk.

Statistical analysis

The normal distribution of the data was verified through the Shapiro-Wilk test. The normally distributed continuous data were presented as mean ± standard deviation with 95% confidence interval (CI), while the categorical data were presented as a percentage over the total.

The rate of athletes with at-risk profile (i.e., number of simultaneous risk factors ≥ 5 in a trial) was computed at IC and MKF for each trial. A multivariate logistic regression model was built to investigate the outcomes associated with the occurrence of the at-risk profile in each movement and frame (four models). In particular, sex (male, female) and limb dominance (i.e., if the movement is performed with dominant or non-dominant limb) were inserted in the model as factors, while age and approaching speed (centre of mass speed calculated at IC) were inserted in the model as covariates. The association was considered statistically significant, and odds ratio (OR) with 95% CI were reported if the likelihood ratio test had $p < 0.05$ to quantify each factor association's strength.

Furthermore, the two-tailed Student's *t*-test within the Statistical Parametric Mapping (SPM1D—spm1d version M.0.4.5 (2017.06.22); http://www.spm1d.org) was used to compare the knee joint moment waveforms between the athletes presenting the at-risk profile and those who did not in each movement and frame. All the statistical analyses were conducted in SPSS (v26, IBM, Armonk, NY, USA).

Results

Overall, 398 movement trials (COD n = 202, DEC n = 196) were collected and further analysed. The remaining 10 trials (2.4%) were excluded due to technical reasons, i.e., error in data acquisition or export in the wearable sensor system. The average approaching speed was 2.8 ± 0.7 [95% CI: 2.7–2.9] m/s for the COD and 1.8 ± 0.6 [95% CI: 1.7–1.9] m/s for the DEC task.

Table 2. Summary of the investigated trials and occurrence of at-risk profile for ACL injury.

Rate of at-risk profile occurrence [a]		
Task, frame	Valid trials	≥ 5 Risk factors, n° (%)
Change of Direction, IC	202	24 (11.9)
Change of Direction, MKF		17 (8.4)
Deceleration, IC	196	24 (12.2)
Deceleration, MKF		19 (9.7)
Overall, IC	398	48 (12.1)
Overall, MKF		36 (9.0)

a = athletes exhibiting at least 5 biomechanical risk factors simultaneously; IC = initial foot contact; MKF = maximum knee flexion angle.

Table 3. Multinomial logistic regression (likelihood ratio test with $p < 0.05$) for the outcomes associated with the occurrence of the biomechanical risk profile for ACL injury.

Outcomes associated with at-risk profile occurrence[a]		
Change of direction, MKF	Odds ratio [95% CI]	*p*-value
Age	0.9 [0.7–1.0]	n.s.
Approaching speed[b]	4.3 [1.6–11.7]	0.004
Sex, Female	4.8 [1.2–19.8]	0.031
Sex, Male[c]	.	.
Limb dominance, 0	0.8 [0.3–2.5]	n.s.
Limb dominance, 1 [c, d]	.	.

a = athletes exhibiting at least 5 biomechanical risk factors simultaneously; b = at initial foot contact; c = reference category; d = movement done with dominant (kicking, 1) or non-dominant (standing, 0) limb; MKF = maximum knee flexion angle.

The injury risk profile (at least 5 simultaneous RFs per trial) was present in between the 9% and 12.1% of the trials (Table 2). Similar rates ($p > 0.05$) were found in terms of motor tasks (COD and DEC) and frames (IC and MKF). Twenty-four athletes (71%) exhibited the injury risk profile in at least one movement. Five athletes (15%) exhibited the risk profile in at least three movements (i.e., one out of four).

The multivariate regression model of the COD, MKF showed statistically significant data fitting (likelihood ratio $p < 0.05$). A higher approaching speed (OR 4.3) and the female sex (OR = 4.8) were significantly associated with the injury risk profile (Table 3).

A statistically significant greater KAM was found in the initial and final parts of the foot strike (15–18%, p = 0.044; and 88–92%, p = 0.046 respectively) in athletes with injury risk profile at the MKF frame of the COD task (Figure 2). Greater KFM and KRM were also found in the 20–75% and 52–81% of the COD task, respectively ($p < 0.001$, Figure 2). Similar results were found for the DEC task (Supplementary Appendix A1).

Discussion and implications

The present study is the first one to investigate the biomechanics of healthy pivoting sport athletes in the light of a multi-joint risk profile for ACL injury. The most important finding was a large occurrence of the at-risk biomechanical profile during the high-dynamics movement investigated, up to 12% of the trials (36–48 out of 398, Table 2).

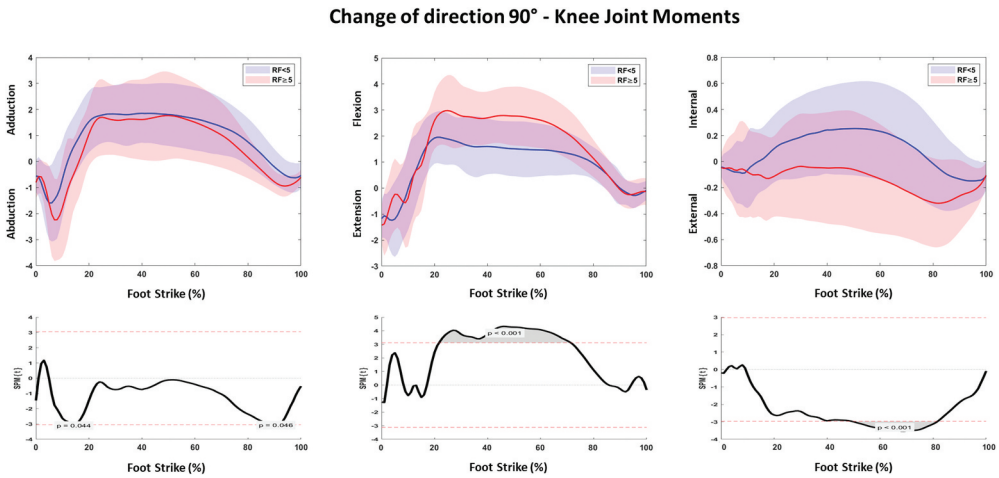

Figure 2. Knee joint moment (N × m/bw) waveform analysis according to the at-risk profile in the maximum knee flexion angle during the Change of Direction task. Data are presented as mean (solid lines) and standard deviation (shaded areas). Top row: the red lines indicated the trials were the at-risk profile occurred (≥5 RF—risk factors), while the blue lines represent the trials were the at-risk profile did not occur (<5 RF); Bottom row: *t*-test comparison within the Statistical Parametric Mapping (SPM1D) for the entire waveform. The grey areas indicate statistical significance ($p < 0.05$).

A limited number of athletes exhibited the profile multiple times; the majority, at least once. Moreover, the at-risk profile was more likely to occur in movements performed with higher approaching speed and by female athletes. Therefore, our hypotheses were confirmed.

The results of the present study suggest that the athletes involved in pivoting sports frequently fall into unsafe motion patterns commonly found in the ACL injury frame. Such a biomechanical profile might increase the risk of knee ligament overload and subsequent ACL rupture, as also suggested by the significant interaction with the knee joint moments in all anatomical planes (Figure 2) (Donelon et al., 2020; Hewett et al., 2015; Sigurðsson et al., 2020). Considering the competitive healthy cohort under investigation, the recurrence of such a biomechanical profile might be surprising. Nonetheless, the body of literature on ACL injury prevention strategies agrees on the fact that limited attention is commonly paid to movement quality in trainings at all levels (Loose et al., 2018; Silvers-Granelli et al., 2017; Thompson et al., 2017), in favour of performance. Where preventative strategies—specifically oriented to movement quality enhancement—have been introduced, reduced occurrence of risk factors for ACL injury has been reported (Silvers-Granelli et al., 2017; Slauterbeck et al., 2019).

However, due to the complex nature of the ACL injury, the most adequate way to assess such risk factors is still under debate. Typically, biomechanical laboratory studies have considered the effect of a limited number of risk factors on the ACL injury risk: most studies focused on dynamic knee valgus (Dempsey et al., 2009; Donelon et al., 2020; Wilczyński et al., 2020), while others investigated the effect of foot position and trunk bending (Hewett et al., 2009; Jamison et al., 2012; Lindblom et al., 2020). Recently, Dix et al. (Dix et al., 2020, 2021) proposed a novel measure of the 'valgus collapse' that simultaneously accounts for the hip internal rotation, hip abduction and knee valgus. The

assessment of multiple concurrent risk factors for ACL injury ('perfect ACL loading storm' (Lloyd, 2021)) has been advocated in contrast to a reductionist approach. The present study is the first to use full-body simultaneous quantitative cut-offs in the assessment of athletes' motion, thus providing a first application of the latter concepts in primary ACL risk assessment. From a clinical perspective, such an approach offers a more comprehensive overview of athletes' movement strategy compared to classical testing and has a practical application in the ACL injury risk screening (Hewett & Bates, 2017).

Higher approaching speed in cut manoeuvre and female sex were significantly associated with the higher occurrence of unsafe patterns in the movement trials. Both aspects are extremely in line with the current literature. First, a 'performance/injury risk' conflict has been recently advocated; Dos'Santos et al. (2021) and McBurnie et al. (2019) depicted greater multiplanar knee joint moments (higher ACL injury risk) in the presence of faster cut manoeuvres (higher performance). Similar to the present study, such results were obtained with a cohort of healthy athletes belonging to a multidirectional sport. When testing athletes for ACL injury risk, a dedicated space is mandatory to allow multidirectional movements at game-like speeds. An ecological environment, including sport-specific external perturbations, is selection of choice (Bolt et al., 2021). Second, higher ACL injury rates commonly reported in female athletes (Ford et al., 2010; Mokhtarzadeh et al., 2017; Prodromos et al., 2007; Silvers-Granelli, 2021) have been ascribed to different aspects, including the biomechanical attitude (greater knee valgus tendency (Ford et al., 2005; Malinzak et al., 2001; Waldén et al., 2011)). The higher occurrence of the risk profile found in the present study also suggests a greater tendency to unsafe patterns in females compared to males. Given the increasing level of competitiveness and the earlier age sport specialisation, a stronger focus on preventative strategies in female athletes' population might be beneficial for ACL injury risk mitigation (Geertsema et al., 2021; Thompson et al., 2017).

The knee joint moment waveform analysis suggested an ACL overloading trend in the presence of the risk profile: both movement tasks exhibited a more pronounced valgus loading in the shock absorption phase and a combination of flexion and external rotation loading in the propulsion phase. Such aspects have been demonstrated to stress the ACL and require a quick and effective muscular counterbalance to maintain the joint stability (Maniar et al., 2019; Oliveira et al., 2014).

To the authors' opinion, the frequent adoption of unsafe biomechanical patterns in combination with external perturbations, e.g., an opponent's deceiving action during a pressing situation, might be the final straw to knee overloading and subsequent ACL rupture (Della Villa et al., 2021; Grassi et al., 2017). Since external perturbations are intrinsic of pivoting (team) sports, a stronger attention should be paid to the athletes' attitude towards unsafe biomechanics. Technological advancements (e.g., the use of wearable technology in ecological environment) and deeper knowledge of the injury mechanisms should serve ACL professionals to recognise and preventatively correct such patterns. The association between unsafe biomechanics and ACL injury risk has been recently confirmed in a longitudinal analysis of elite soccer players that identified the non-contact injury mechanism as the strongest significant predictor of ACL re-rupture (HR = 7.16, Della Villa et al. (2021)). Thus, the adoption of large biomechanical screenings in young sporty population is essential for two main

reasons: increase the awareness on the risks derived from unsafe movements and identify the athletes at higher risk (Benjaminse & Verhagen, 2021). The latter might be selected to receive dedicated attention and preventative neuromuscular training (Akbari et al., 2020).

The present study has several limitations. First, no previous studies (either in-lab or on-field) investigated a full-body biomechanical profile with quantitative cut-offs. Thus, it is difficult to compare the results of the present study with the current literature. Moreover, our population did not include ACL-reconstructed athletes. Therefore, no sensitivity analysis of such a biomechanical profile could be conducted. Future studies might investigate whether such a pattern is more frequent in ACL-reconstructed athletes and useful in the prediction of primary ACL injury through prospective studies. Second, the cut-off values used for the biomechanical profile were adapted from the literature for the sagittal plane kinematics (ankle, knee, hip flexion) and form our cohort distribution for the frontal and transverse plane kinematics. Thus, the latter cut-offs were not referring to an injury frame nor representing aberrant biomechanics. However, a robust statistical method such as interquartile range was adopted for a cohort that well represented a typical healthy population involved in pivoting sports: all competitive athletes, equally distributed male and female, with no history of musculoskeletal injuries, not belonging to dedicated prevention programmes. Moreover, the biomechanical profile was based on a large number of risk factors (≥5), thus reducing the risk of bias for single parameters. The number of risk factors set out to define a movement trial as 'at-risk' was arbitrary. However, such a number meant the involvement of risk factors belonging to multiple joint and multiple categories (Table 1). Moreover, as confirmed by the significant association with the knee joint moments, the chosen number of factors increased the risk of knee overload. Third, only two tasks were investigated. The use of multi-task batteries including single- and double-leg landings and unanticipated movements might help understanding if the occurrence of the biomechanical profile is occasional or recurrent. However, the movements investigated resembled typical situational patterns for ACL injury (Della Villa et al., 2020) and athletes' focus was on movement performance (they were asked to sprint at maximum speed), as in field condition. Lastly, the motion analysis was conducted by means of a wearable inertial sensor system instead of gold standard marker-based motion capture. However, the former has been extensively validated for dynamic movements and, recently, for the investigation of ACL injury mechanisms (Di Paolo et al., 2021; Poitras et al., 2019). Tools capable of quick biomechanical assessments with limited encumbrance, e.g., wearables and markerless technology, are crucial to endorse the adoption of large screenings in young sports population.

Conclusions

The present study underlined a large occurrence of typical biomechanical profile for ACL injury in young healthy athletes during the execution of complex dynamic movements. Large screening in the young population is advisable for the early identification of at-risk athletes and the adoption of preventative training strategies focused on the mitigation of unsafe biomechanical patterns.

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